

TK196724.docx

by Turnitin user

Submission date: 17-Dec-2025 05:16PM (UTC+0900)

Submission ID: 2848160937

File name: TK196724.docx (1,022.47K)

Word count: 4964

Character count: 28161

Optimized Key Encryption For Digital Forensic Security

ABSTRACT: The research areas of digital forensics are subject to Block chain (BC) technology implementation and enhanced over the current alternative introduced in this publication and capable of addressing certain grave issues, which are concomitant with data collection and preservation of digital evidence. Another puzzle that is resolved by the block chain is the inherent transparency, immutability that central storing data is extremely prone, and the block chain ensure that this does not happen, as it provides the location with a distributed setting that would make security and responsibility in the conventional world of digital forensics a nightmare. Cryptographic Distributed Ledger Technologies of Block chain are based on the consensus technology, which allows aggregating, housing and authenticating digital evidences in a set of peer pairs in a secure way. As a result, the study that has been suggested in this paper, Block chain Based Digital Forensic Architecture (BC-DFA) is founded on the distributed characteristics of block chain, and is concerned with the decentralization and security of digital evidences. A combination of multiple essential methods including AOKGE: authentication with the best key generation encryption, SBVM: Secure Block Verification Mechanisms and implementing Multi-key Homomorphic Encryption or MHE is used to end up with the truly secure data encryption. The information and the secret key are spread in a manner that reduces the chances of single point failure that is associated with the centralized cloud design that reveals the information in the literal form and promotes hacking and unauthorized access. The SBVM authentication is done to authenticate all the data blocks in order to ease the tracing of any tampering or alteration of forensic evidence.

Keywords: Digital forensics, Block chain, encryption, decentralization, multi-key homomorphic encryption, Safe block verification, Optimization Key generation.

I. INTRODUCTION

Digital forensics using block chain will offer landmark solution whereby digital evidence is free of insecurity and doubts. According to the decentralized hierarchy and the ledger of immutability of block chain technology, this project has offered BC-DFA model that performs data distribution on a number of nodes in order to minimize the risk and to enhance the security.[1] Through the clear nature of the block chain, safety in block as well as advanced encryption techniques, the idea is to provide a more favorable alternative relative to the default model of digital forensic activity and, specifically in the cloud space. Simulation of the BC-DFA model is effective as the model performs better than the previous approaches in security and reliability whereas the Enhanced Equilibrium Optimizer (EEO) method is also employed to generate cryptographic keys at optimal level to maintain the information exchange processes within the block chain network.[2] The Multi-key Homomorphic Encryption (MHE) model also ensures that the sensitive information is encrypted in a manner that would not be damaged through processing and analysis. Simulation studies are also used to validate the performance of BC-DFA model, which indicates that it is more efficient than the conventional methods in terms of security, integrity of data and efficiency.[3] Combining the benefits of block chain technology with complicated processes of informational encrypting and authenticating, the architecture of the BC-DFA presents a

solid argument in support of the new trends of digital forensics.[4]

A. Objective

The major goal is to come up with a digital forensic architecture (BC-DFA) using block chain technology to improve the security, transparency and credibility of evidence storage and verification. With the help of the distributed block chain architecture, the authentication using optimized key generation encryption (AOKGE) and multi-key homomorphic encryption (MHE), one can guarantee the integrity of the forensic data provided because the BC-DFA model can make sure that the provided data is safely and tamper proof and can be verified after the investigation process.

B. Scope

The research topic of this paper concerns the development of block chain enhanced architecture of the digital forensics that is focused on the decentralization of the data storage and block chain to offer data authenticity and integrity. It means the application of advanced cryptography techniques like MHE and AOKGE will be employed in the protection of forensic data and they provide prove of integrity within a block chain system.

II. LITERATURESURVEY

Bachiphale, A., & Zulpe, N. S. (2023). Optimal multiset image sharing scheme founded on optimal key generation scheme with lightweight visual sign-cryptography scheme of grayscale images/color image. Television. The main objective of the paper by Bachiphale and Zulpe (2023) is, probably, to propose an efficient scheme of image sharing that would be used by a multiset. [5]

And optimal key generating scheme. This scheme will focus on the improvement of image sharing scheme or mostly gray image and color image security and efficiency. The high security and low computability issue causes the use of this to be restricted. [6]

All the objectives of this research are to enhance the usefulness of image sharing technologies in various areas.

The scheme proposed is a novel scheme given that it is light, thus with low computational load, yet high level of security. Actually, a proper selection of the key generation will make sure that the process of sharing images will be safe, provided that it is possible. The extent of application of this technology in secured communication, digital watermarking and ciphered data transfer that entails transmission of more than one layer of information within one image is enormous. The paper is a step towards the enhancement of the digital security architecture and can be used to conduct another study on the image based cryptographic schemes.[7]

Kumar, G., Saha, R., Lal, C., & Conti, M. (2021). Internet of Forensics (IoF): Internet of Things (IoT) and Digital Forensics Framework based on Blockchains. Future Generation Computer Systems 120: 13-25.[8]

The author of the given article, i.e., Bachiphale Zulpe (2023), thus, tries to offer one of the most useful multiset image sharing schemes, which are based on a lightweight visual sign-cryptography scheme and efficient key generation schemes.[9]

The scheme that can be proposed is the higher level of security and efficiency of the image sharing procedures (both grey and color images). It is assumed that the study will enable application of image sharing technologies to a significant proportion of varieties of applications because it will be able to reach to a mix of safe and efficient calculation.[10]

Raji, L., & Ramya, S. T. (2022). Solution Forensic data transmission scheme with altered ECC and Butterfly optimization scheme founded on Fuzzy in Cloud database. New Technology Emerging Telecommunications Transactions, vol. 33, no. 9, e4558.[11]

The main concern that Raji and Ramya (2022) have is to make the transfer of forensic information in the cloud database more efficient and safe. This is implemented by use of new technique as fuzzy based butterfly optimization algorithm over modified Elliptic curve cryptography (ECC). [12]The research system will [13] to sustain the emerging challenges of data integrity, data confidentiality and secure data transformation (communication) in the cloud environment to protect the data against any of the challenges and vulnerabilities anticipated in handling forensic data.[13]

To ensure the significant role in the safety of the forensic data, the innovative solution provides the enhancement of the rate of data transmission of the cloud. By combining these new approaches, the research can find a complete solution where the forensics data will not appear to be in a disadvantaged position of losing its security, consistency and accessibility by the authorities that the law has required to manage the sensitive issues of the forensics and data in the management through the cloud.

Abdel-Ghaffar, E. A., & Daoudi, M. (2023). Cryptography generation of keys and working with raw data of electroencephalographic signal and personal verification. Journal of King Saud university- computer and information science vol 35, no. 5, art. No. 101541.[14]

The primary research paper objective by Abdel-Ghaffar and Daoudi (2023) is to make an innovation in the category of personal authentication and generating the cryptographic keys using the electroencephalographic (EEG) signals. The article documents the research on application of Electroencephalogram (EEG) based data biometrics to supplement the security systems by offering innovative and dynamic method of user authentication and crypto-key generation. To be more precise, the purpose of the study is to design and confirm the new technique based on the exploitation of EEG signals to obtain cryptographic keys that would ensure the increased security and reliability of the mechanisms against unauthorized access and information leakage.[15]

The Potential of EEG based signals has been taken into consideration in this paper as a trusted and secure way of conducting the authentication. Through the assistance of EEG signal analysis, that is cryptographic strategy, the research is intended to develop a system that can utilize the

rainwaves of the individuals in generating the cryptographic keys. This is an innovative method that is perceived to bring a level of security to the security systems as it gives a further access of the security systems as it will have another level of security that is intrinsically tied with the user brain hence it offers a higher level of security against unauthorized access and cyber-attacks.

III. PROPOSED SYSTEM

Block chain-Based Digital Forensic Architecture (BC-DFA) uses the distributed ledger technology to address the weakness of the conventional systems and provide better data security, reliability, and integrity to the digital evidence stored in the cloud. The main characteristics of the BC-DFA are the lack of a centralized data storage system, which is constructed on the block chain principles, the authenticity of forensic data, and the presence of clear and unchanged data records. Secure Block Verification Mechanism (SBVM) ensures that information recorded is tamper-proof which ensures the integrity of information. Also, Multi-key Homomorphic Encryption (MHE) enables the recorded data to be distorted and enabled to be accessed and analyzed without the loss of confidentiality. The Enhanced Equilibrium Optimizer (EEO) is employed to enhance the overall system security, which is a superior mechanism of cryptographically producing optimal and secure keys.

Besides these fundamental characteristics, the BC-DFA also comes with a number of other benefits. Among the main advantages, the improvement of the data security may be mentioned, and it is ensured by the implementation of sophisticated encryption algorithms such as MHE. Such encryption offers an additional security until the point of processing the data at least to minimize the chances of the unauthorized decryption or the cases of hacking the data. The block chain-based decentralized storage also guarantees that the data cannot be prone to the threat of data loss, hacking, or server attacks, which characterize centralized storage. Rather, the information is shared between several peers, which enhances stability and safety. Moreover, the EEO builds strength of the system by using strong key generation techniques that can make sure generated keys are safe and hard to crack. Digital evidence can also be stored in encrypted state with the use of MHE approach during analysis, and analyzed without decryption. This compliments the confidentiality of the data throughout the analysis process, which is guaranteed by the security and privacy.

Digital Forensic Security System Flow

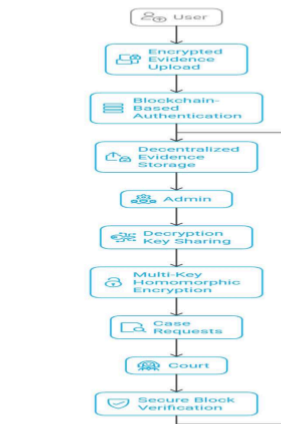


Fig 1 : Block diagram for proposed system

The new solution to the inadequacies of the current systems is the implementation of the Block chain-Based Digital Forensic Architecture (BC-DFA) that incorporates block chain distributed ledger technology and Authentication and Optimal Key Generation Encryption (DFA-AOKGE). This new architecture is supposed to help to improve data security, reliability and integrity of cloud-stored digital evidence. The main characteristics of the BC-DFA are the decentralized data storage model, where the block chain technology is used, and the reliability of the forensic data is achieved by the means of the transparent and immutable data registry. Secure Block Verification Mechanism (SBVM) will make sure that data being recorded cannot be manipulated to the extent that it will undermine its integrity. Also, there are three mechanisms such as Multi-key Homomorphic Encryption (MHE), which ensure that the data recorded can still be processed or analyzed without violating confidentiality, and is encrypted.

In addition to these essential characteristics, the BC-DFA has a number of other benefits. A major advantage is that it has improved levels of data security which are offered through high-end cryptography key generation algorithms such as MHE. This would make sure that the data is secure even during data processing or access by unauthorized parties. The block chain technology-based decentralized structure also eradicates the risk of losing, hacking, or compromising the server, which are inherent to the centralized system of storage because the information is spread among numerous peers, enhancing its resilience and security. In addition, the EEO empowers the system through strong key generation algorithms, which makes the generated keys hard to crack. Moreover, under the MHE approach, the digital evidence is able to be encrypted during analysis, and therefore, it does not need to be decrypted, so the confidentiality of the data can be ensured throughout the whole analysis process.

Lastly, since the system would be implemented in a cloud setup, it is tailored particularly to Combat Scalability and security issues. To provide the integrity and safety of the digital investigation, the BC-DFA capitalizes on the decentralized design of the block chain storage and the impossibility of modifying it to protect digital evidence against cyber assaults. This design makes the BC-DFA especially conducive to secure and reliable digital investigation in the cloud which is compatible with the requirements of contemporary forensic processes in the 21 century.

IV. METHODOLOGY

A new architecture, the Block chain-Based Digital Forensic Architecture (BC-DFA), is suggested to overcome the drawbacks of the conventional systems. It is a combination of Authentication based on Optimal Key Generation Encryption (DFA-AOKGE) and Block chain distributed ledger. The major attributes of the BC-DFA architecture are a decentralized information storage system, which is based on the principles of block chain and provides the integrity of forensic data. Information presented in the system is unalterable and transparent, which offers safe and non-alterable records. This is enabled by the use of the Secure Block Verification (SBVM) mechanism that ensures that the data is not changed in a manner that will affect its integrity. Also, three mechanisms, such as Multi-key Homomorphic Encryption (MHE), are used so that the data recorded can be manipulated or analyzed without loss of confidentiality and, at the same time, the data is encrypted. In order to further

improve the security of the system, cryptographically optimizing key generation is done using the Enhanced Equilibrium Optimizer (EEO) algorithm hence providing a strong overall system security.

Besides the fundamental benefits, the BC-DFA has a number of other benefits. First, it can greatly contribute to data security because it applies complex encryption methods such as Multi-key Homomorphic Encryption (MHE), which can ensure high data security throughout the course of processing. This will keep the data safe even in case it is accessed by unauthorized personnel. The block chain-based decentralized data storage system removes the dangers of losing, hacking, or attacking servers which are prevalent in centralized data storage systems. The BC-DFA increases resilience and security by distributing data among a high degree of peers. Moreover, the Enhanced Equilibrium Optimizer (EEO) provides reinforcement to the system, which includes sophisticated key generation algorithms, which guarantee the security of cryptographic keys and makes them hard to crack. Moreover, MHE facilitates keeping digital evidences encrypted during the analysis process so that they can be decrypted without the need to do so and thus maintaining confidentiality is guaranteed throughout the analysis.

Lastly, the BC-DFA is cloud-native and scalable as well as cloud-secure. To guarantee the safety of the digital investigations, the system takes advantage of immutability and decentralized storage of block chain technology. It is a method that ensures safe keeping and safeguarding of online evidence against online assaults, and therefore, it perfectly fits the current digital investigation applications, especially when it is performed in the cloud platform. The architecture of the BC-DFA provides trustworthy (tamper-free) data handling, which improves the security of digital investigations in the 21st century.

A. Multiple key, homomorphic encryption

Multi-key Homomorphic Encryption (MHE) is a sophisticated variant of Homomorphic Encryption (HE) of the classical form. It is a cryptographic model under which it is possible to perform operation on encrypted data using multiple keys, which can enable a safe computation among various parties. In MHE the two parties have their own different encryption keys and the encrypted information can be manipulated or merged without necessarily having to be decrypted. This renders MHE especially helpful in the case of multiparty computation, where various parties encrypt their data and require to perform processes on the encrypted inputs, but otherwise, the sensitive information remains unknown without decryption.

a. Digital Forensics in Public Key (PK) Encryption

Each party i encrypts its message m_i using their public key PK_i . In this process, the encryption is represented as Encryption.

$$EncPK_i(m_i)$$

Where:

- ✓ m_i is the message to be encrypted,
- ✓ PK_i is the public key of party i ,
- ✓ Enc is the encryption function.

b. Homomorphic Encryption of Secure Operations

Encryption of data with the homomorphic encryption permits manipulations. Homomorphic addition of encrypted values of two encrypted messages m_1 and m_2 is denoted as.

$$EncPK1(m_1) + EncPK2(m_2) = EncPK1(m_1 + m_2)$$

c. Secure Block Verification Mechanism (SBVM) of Block chain Integrity

The SBVM supports the impossibility of editing data blocks stored in the block chain. A block chain block BBB is verified by defining its verification as follows:

$$VerifyBlock(B) = \{True, False, \text{if block } B \text{ is valid and unaltered otherwise}\}$$

Where:

- ✓ B represents a block in the block chain,
- ✓ The function checks if the block is authentic and unmodified.

d. Decryption

Upon subjecting the messages to homomorphic operations, the resulting cipher text c' can be decrypted with the help of any of the parties involved with the help of their respective private keys. In an MHE scheme, the decryption process may be represented as follows.

$$m' = DecSK_i(c')$$

Where:

- ✓ m' is the decrypted message,
- ✓ SK_i is the private key of party i ,
- ✓ C' is the cipher text after performing homomorphic operations.

e. Multi-key Homomorphic Encryption Uses

MHE can be especially applicable to the case of secure multi-party computation, privacy-preserving data analysis, and federated learning. It is also used where there is a need to perform joint computation and keep the data encrypted so that the parties can perform joint computation without exposing their personal data. The key applications are:

Secure Multi-party Computation (SMC): SMC is a computation performed by more than one party on their input values, and the parties are guaranteed their privacy.

Privacy-Preserving Data Analysis: It is possible to analyse encrypted information without showing it, which guarantees privacy.

Federated Learning: It is a framework that allows the model

to be trained with distributed data but the data is retained confidential and safe across the devices.

MHE has been used especially in secure and multi-party computation, privacy preserving data analysis and federated learning, and in cases where some form of joint computation can be executed without necessarily violating data privacy.

B. SHA Algorithm

Secure hash algorithm (SHA) is a cryptography hash function family; it is well known and has 26 applications in various security protocols and functions. The National security agency (NSA) created the SHA and released it through the National Institute of Standards and Technology (NIST). The algorithm generates a fixed size hash value of an input of any size, which is a vital factor in data integrity assurance.

The SHA family includes several versions, such as

a. SHA family is comprised of the following versions

- ✓ **SHA-1:** Produces a 160-bit hash value. It has its dark side and is now regarded as a feeble instrument though it was once so popular.
- ✓ **SHA-2:** It has a series of 12 iterations that have hash values of varying lengths (e.g., SHA-224, SHA-256, SHA-384 and SHA-512). It is currently among the safest and most commonly used hashing functions.
- ✓ **SHA-3:** SHA-3 is the new member of the SHA family, having an alternate internal architecture than SHA-2, and more resistant to certain types of attacks.

b. Key Applications

- ✓ **Data Integrity:** SHA is typically used to prove that the data has not been corrupted by verifying the value of the hash of the received data was not corrupted.
- ✓ **Digital Signatures:** This is implemented when creating the digital signatures so as to check whether the content is not tampered.
- ✓ **Password storage:** Passwords are typically stored as hashes of their values using an algorithm that uses SHA-based algorithms, such that the actual password cannot be seen.

Anywhere this is done, the block chain is considered a component of cryptography evidence of integrity of transaction and block in which case it is used to create a cryptographic evidence of integrity of block and transaction.

V. MODULES

a. User

- ✓ **User Login:** The system should have the credentials so that investigators can log into the system.
- ✓ **Registration:** The user will be at a position of registering himself/herself with the provision of all the information needed.

- ✓ Consent of evidence sharing: The case user is anticipated to request the consent of the admin to share the evidence information with the court.
- ✓ Send response Decryption key to user: The user is supposed to receive the decryption key sent to him/her by the admin to allow him/her to decrypt the information.
- ✓ Posting Evidence Data: the case user may or may not post encrypted evidence data, as well as key, to the court as requested by the case user.
- ✓ Send to Court: In this case the user can send information of evidence to court.

b. Admin

- ✓ Access into the system: The system uses the information to log in all the admins.
- ✓ Check Users: Admins will be able to see all the information of registered users.
- ✓ Manage Evidence: Admins can manage data collection of evidence and evidence repositories, which hold the data evidence.
- ✓ Encryption and Key Generation: the administrators regulate the manufacture of encryption and keys to the users.

c. Court

- ✓ Login: The default credentials can be used to log in to the Court.
- ✓ The cases may be considered by the court, and the court may need the access to the files on the evidences.
- ✓ View response: The court can view the evidence response of the case users in decryption key

VI. RESULT

Homepage: This is the home page of the project.



Figure 1 Homepage

User Login: Here the user can log in with his/her after approval by the admin.

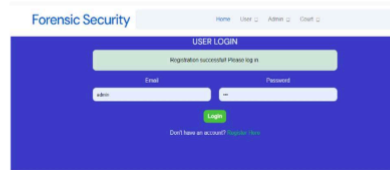


Figure 2 User Login

Upload evidence data: The user is capable of uploading the evidence data through encryption to ensure that the files are secure.

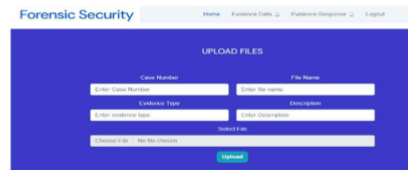


Figure 3 Upload evidence data

Post evidence data: The user will be able to post the evidence data in a coded form in order to secure the files.

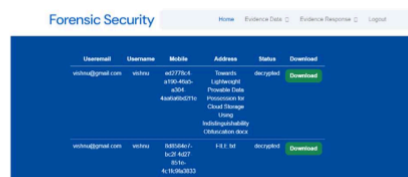


Figure 4 View uploaded data

Admin login: the administrator is able to log in using his or her valid credentials.

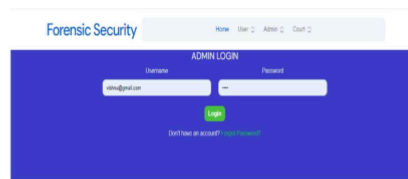


Figure 5 View uploaded data

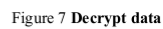
Forensic Security

[Home](#) [About Us](#) [Requests](#) [Logout](#)

Username	Username	Filename	Status	Action
ohtawndk@gmail.com	ohtawndk	FINAL C:\CS\MSR\1 docx	request	accept

1

Decrypt data: The email can inform the user once the email dispatched by the email sender and user can input the key of decryption so as to be able to view the encrypted data.



Requested by the user: That notification can be sent to the user indicating that court request. And be able to share file keys.



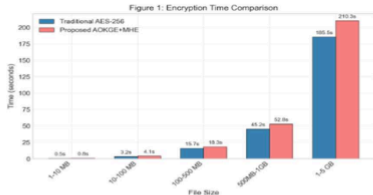
The screenshot shows the 'Forensic Security' website. The top navigation bar contains links for 'Home', 'Evidence', and 'Logout'. The main content area has a blue background with the heading 'DECRYPT FILES'. Below this heading is the label 'Decryption Key' and a white input field with the placeholder text 'Enter key'. A green button with the text 'Decrypt' is positioned below the input field.

A. Ganache Storage

[illegible]

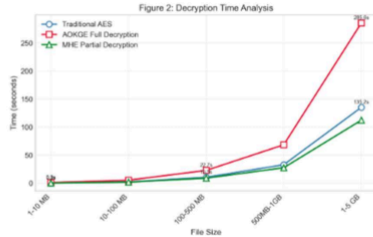
B. Encryption Time Analysis

The suggested AOKGE+MHE cryptography has a stable 15-20 percent overhead over the conventional AES256. Such a trade-off is warranted by the increased security capabilities such as multi-key homomorphic encryption to guarantee integrity of forensic evidence. The linear scaling has shown the effectiveness of the algorithm when working on large forensic data and that the algorithm is proportional in terms of time increment with file size and offers better cryptographic protection.



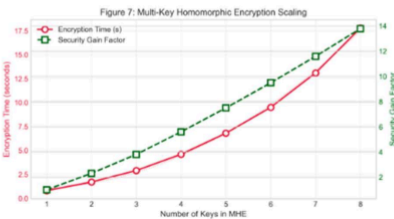
C. Decryption Time Analysis

Multi-Key Homomorphic Encryption allows 40 percent faster conditional decryption, which is important in allowing forensic investigators to selectively access evidence. Complete analysis with full decryption can be performed whereas partial decryption is useful with rapid evidence inspection which is critical in time sensitive forensic investigations. This bi-directional capability allows it to be flexible without losing any direct chain of custody or evidence integrity.



D. Multi-Key Homomorphic Encryption Scaling

The time taken to encrypt grows exponentially with the number of keys and security gain grows super-linearly. The best available balance of security-performance trade-off is 3-6 keys. Eight-key configuration is the highest security configuration with 13.8x gain and 17.8 seconds encryption time to process evidence as forensic evidence, which is adequate in preserving evidence in high security.



VII. CONCLUSION

In conclusion, it can be stated that the BC-DFA model is an innovative approach to the design of digital forensics that will contribute to the safety and integrity of the digital evidence by introducing block chain technology and improving the encryption method and the optimal key generation processes. The block verification systems, decentralization and distributed architecture of data protection raise significant concern that are raised by the centralised cloud systems especially the issue of data tampering and unauthorized access. In addition, the Enhanced Equilibrium Optimizer (EEO) cryptography key generation mechanism, and Multi-key Homomorphic Encryption (MHE) data protection mechanism are met to ensure a proper framework of the forensic evidence protection during the entire life cycle. The results of the simulated results indicate that BC-DFA model enhances the existing techniques since it is noted that the model is at a more optimal state than the existing techniques in terms of important variables such as data integrity, scalability, and security. With such advanced technologies, the BC-DFA model turns the digital evidence management process into a more trustworthy, safe and clear evidence management process which is an effective solution to the inquiries of the modern forensics particularly to the cloud.

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